



$$\begin{array}{c}
p, \pi, \rho \in \mathcal{P}_2^r, G \text{ is } \lambda\text{-convex a.g.g.}, \\
G(\pi) - G(\rho) \geq \langle \nabla_{\mathcal{W}_2} G(\rho) \circ T_p^\rho, T_p^\pi - T_p^\rho \rangle_p + \frac{\lambda}{2} \mathcal{W}_2^2(\pi, \rho) \\
\hline
\pi, \rho \in \mathbb{R}^d, g \text{ is } \lambda\text{-convex}, \\
g(\pi) - g(\rho) \geq \langle \nabla g(\rho), \pi - \rho \rangle + \frac{\lambda}{2} \|\pi - \rho\|^2
\end{array}$$

Figure 2: The monotonicity of a.g.g.-convex G in $\mathcal{P}_2$ proved in Lemma 4.1, as an analog to strong convexity in vector space. We remark that in the usual vector space, the convexity definition does not involve a third vector, since the inner product is uniform; while in probability space, inner product is defined at tangent space associated with p . The dotted line indicates the general geodesic between ρ and π , see the definitions in Section 2.2.

empirically shown in [74] that the inversion error computed on testing samples (though all the N blocks) can be made small towards the floating-point precision in the neural-ODE model.

The inversion error objective (33) resembles the “cycle consistency” loss in Cycle-GAN [76]. Our theory in Section 5.2 suggests that keeping the inversion error small is crucial for the success of generating a close-to-data distribution $\tilde{q}_0$ in the reverse process. Thus, there may be benefits by introducing the objective (33) as a regularization in the training of flow-based models, similar to the cycle consistency mechanisms in the GAN literature.

4 Convergence of forward process

The current paper mainly concerns the application to flow generative networks where G is the KL divergence. In this section, we prove the exponentially fast convergence of the forward process, which applies to potentially a more general class of G as long as the subdifferential calculus in $\mathcal{P}_2$ can be conducted (see Section 10.1 of [7]) and G is strongly convex along generalized geodesics (a.g.g., see Definition 2.3), which may be of independent interest.

We will revisit the KL divergence G as a special case and prove the generation guarantee of the reverse process in Section 5. All proofs and technical lemmas are provided in Appendix B.

4.1 Conditions on G and V

We introduce the more general condition of G needed by the forward process convergence.

Assumption 3 (General condition of G). $G : \mathcal{P}_2 \rightarrow (-\infty, +\infty]$ is lower semi-continuous, $\text{Dom}(G) \subset \mathcal{P}_2^r$; G is λ -convex a.g.g. in $\mathcal{P}_2$.

The first part of Assumption 3 ensures that the strong subdifferential $\partial_{\mathcal{W}_2} G(\rho)$ can be defined, see Definition 2.2. The strong convexity of G is used to prove the exponential convergence of the (approximated) $\mathcal{W}_2$ -proximal Gradient Descent in the forward process.

Next, we show that the KL divergence G satisfies the general condition under certain general conditions of the potential function V plus its strong convexity. We also introduce an upper bound of λ due to a rescaling argument.

KL divergence and f -divergence G . Recall that the KL divergence $G(\rho) = \mathcal{H}(\rho) + \mathcal{E}(\rho)$ as defined in (16), where $\mathcal{E}(\rho) = \int V \rho$ involves the potential function V of the equilibrium density. We introduce the following assumption on V :

Assumption 4 (Condition of V the potential of q). *The potential function $V : \mathbb{R}^d \rightarrow (-\infty, \infty]$ is proper, lower semi-continuous, and $V^- := \max\{-V, 0\}$ is bounded; $V(x)$ is λ -strongly convex on $\mathbb{R}^d$, and $q \propto e^{-V}$ is in $\mathcal{P}_2^r$.*

The first part of Assumption 4 is for the sub-differential calculus of $\mathcal{E}(\rho) = \int V \rho$ in $\mathcal{P}_2$. The λ -strong convexity is used to make $\mathcal{E}(\rho)$ (and subsequently $G(\rho)$) λ -convex a.g.g. Under such condition of V , the KL divergence $G(\rho)$ satisfies Assumption 3, which is verified in Lemma B.3. Thus our result applies to the KL divergence as being used in the JKO flow model. For the important case when q is standard normal, $V(x) = \|x\|^2/2$, and $\lambda = 1$.

More generally, consider the f -divergence $G(\rho) = D_f(\rho||q) := \int f(\rho/q)q$, $f : (0, \infty) \rightarrow \mathbb{R}$ being convex, lower-semicontinuous and $f(1) = 0$. KL divergence corresponds to a special case where $f(x) = x \log x$. Under Assumption 4, we have $q \propto e^{-V}$ and is log-concave, and then $G(\rho)$ is convex a.g.g. in $\mathcal{P}_2^r$ (under additional technical conditions on f) [7, Theorem 9.4.12]. We think it is possible to establish the λ -convexity a.g.g. of f -divergence G for a certain class of f , and details are postponed here.

Positive and bounded λ . Note that for strongly convex V we can use a scaling argument to make λ bounded to be $O(1)$ without loss of generality. Specifically, for V that is λ -convex on $\mathbb{R}^d$, the function $x \mapsto V(ax)$ for $a > 0$ is $(a^2\lambda)$ -convex. This means that for q that has a strongly convex V as the potential function, one can rescale samples from q to make V strongly convex with $\lambda \leq 1$. In the case where G is KL divergence, the λ -convexity of G has the same λ as that of V . Thus, for the general G we assume its λ is also bounded by 1.

Assumption 5 (λ bounded). *In Assumptions 3 and 4, $0 < \lambda \leq 1$.*

Our technique can potentially extend to analyze the $\lambda = 0$ case, where an algebraic $O(1/n)$ convergence rate is expected instead of the exponential rate proved in Theorem 4.3. For the application to flow-based generative model, one would need the equilibrium density $q \propto e^{-V}$ convenient to sample from, and thus the normal density (corresponding to $V(x) = \|x\|^2/2$) is the most common choice and the other choices usually render $\lambda > 0$ (to enable fast sampling of the starting distribution). We thus leave the $\lambda = 0$ case to future work.

4.2 Evolution Variational Inequality and convergence of the forward process

The a.g.g. λ -convexity of G leads to the following lemma, which is important for our analysis. All proofs in this section are provided in Appendix B.

Lemma 4.1 (Monotonicity of G). *Let $p, \rho \in \mathcal{P}_2^r$, $\pi \in \mathcal{P}_2$, and denote by T_p^ρ and T_p^π the OT maps from p to ρ and to π respectively. Suppose G satisfies Assumption 3, then for any $\eta \in \partial_{\mathcal{W}_2} G(\rho)$,*

$$G(\pi) - G(\rho) \geq \langle \eta \circ T_p^\rho, T_p^\pi - T_p^\rho \rangle_p + \frac{\lambda}{2} \mathcal{W}_2^2(\pi, \rho).$$

The relationship among p, ρ, π is illustrated in Figure 2, which also includes an analog to the strong-convex function in Euclidean space. This lemma extends Lemma 4 in [63] and originally the argument in Section 10.1.1.B of [7]. We include a proof in Appendix B for completeness.

Based on the monotonicity lemma and the condition of small strong subdifferential ξ_{n+1} in Assumption 1, we are ready to derive the discrete-time *Evolution Variational Inequality* (EVI) [7, Chapter 4] for the (approximate) JKO scheme.

Lemma 4.2 (EVI for approximate JKO step). *Given $\pi \in \mathcal{P}_2$, suppose G satisfies Assumption 3 with $\lambda \in (0, 1]$, and $0 < \gamma < 2$. If $p_0 \in \mathcal{P}_2^r$, and Assumption 1 holds for $n = 0, 1, \dots$, then for all n ,*

$$\begin{aligned} & \left(1 + \frac{\gamma\lambda}{2}\right) \mathcal{W}_2^2(p_{n+1}, \pi) + 2\gamma (G(p_{n+1}) - G(\pi)) \\ & \leq \mathcal{W}_2^2(p_n, \pi) + \frac{2\gamma}{\lambda} \varepsilon^2. \end{aligned} \quad (34)$$

The condition of $\lambda \leq 1$ and $\gamma < 2$ can be replaced with other constants, and our analysis will give similar results. Specifically, the upper bound 1 of λ in Assumption 5 is a generic choice and we keep it, then replacing the requirement of $\gamma < 2$ with $\gamma < \gamma_{\max}$ for another $\gamma_{\max} > 1$ will only affect the constant in the final bound and does not affect the order. For exhibition simplicity, we give our analysis under $\lambda \leq 1$ and $\gamma < 2$ without loss of generality. We also provide some rationale for upper bounding the step size γ motivated by practice: First, it has been empirically observed that successful computation of the JKO flow model in practice needs the step size not to exceed a certain maximum value, which is an algorithmic parameter [74]. Setting the step size too large may lead to difficulty in training the Residual blocks as well as in maintaining small inversion errors. Meanwhile, from the formulation of the JKO scheme, it can be seen that for large γ , the proximal GD in (17) approaches the global minimization of $G(\rho)$, which asks for the flow to transport from the current density to the target density q in one step. Though the proximal GD (as a backward Euler scheme) does not impose a step-size constraint, the optimization problem (17) is, in principle, easier with a small (but no need to converge to zero) step size.

The EVI directly leads to the N -step convergence of the forward process, which achieves $O(\varepsilon)$ $\mathcal{W}_2$ -error and $O(\varepsilon^2)$ gap from the optimal objective value in $N \lesssim \log(1/\varepsilon)$ JKO steps.

Theorem 4.3 (Convergence of forward process). *Suppose $q \in \mathcal{P}_2$ is the global minimum of G , and the other assumptions are the same as Lemma 4.2, then for $n = 1, 2, \dots$,*

$$\mathcal{W}_2^2(p_n, q) \leq \left(1 + \frac{\gamma\lambda}{2}\right)^{-n} \mathcal{W}_2^2(p_0, q) + \frac{4\varepsilon^2}{\lambda^2}. \quad (35)$$

In particular, if

$$n \geq \frac{8}{\gamma\lambda} (\log \mathcal{W}_2(p_0, q) + \log(\lambda/\varepsilon)), \quad (36)$$

then

$$\mathcal{W}_2(p_n, q) \leq \sqrt{5} \frac{\varepsilon}{\lambda}, \quad G(p_{n+1}) - G(q) \leq \frac{9}{2\gamma} \left(\frac{\varepsilon}{\lambda}\right)^2. \quad (37)$$

Remark 4.1 (Comparison to [63]). The convergence rates of Wasserstein proximal GD were previously studied in [63], and our proof techniques, namely the monotonicity of G plus discrete-time EVI, are similar to the analysis therein. However, the setups differ in several aspects: first, we consider the “fully-backward” proximal GD, i.e., the JKO scheme, while [63] focuses on the forward-backward scheme to minimize G having a decomposed form $\mathcal{E} + \mathcal{H}$ (which does cover the KL convergence as a special case). Second, [63] assumed the exact solution of the proximal step while our analysis takes into account the error ε in the first-order condition Assumption 1, which is more realistic for neural network-based learning. At last, [63] assumed L -smoothness of V , namely ∇V is L -Lipschitz, and step size $\gamma < 1/L$, as a result of the forward step in the splitting scheme, which is not needed in our fully backward scheme. The motivation for [63] is to understand the discretized Wasserstein gradient flow and to recover the same convergence rates as the (forward-backward) proximal GD in the vector space. Our analysis is motivated by the JKO flow network, and the forward process convergence is an intermediate result to prove the generation guarantee of the reverse process.